

FJAERLAND BREMUSEET, NorwayWMO: **013320**Lat: **61.4231N**Lon: **6.7642E**Elev: **10**StdP: **14.69**Time Zone: **1.00 (EUC)**Period: **06-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-0.2	4.7	-4.0	4.4	-0.1	0.8	5.7	5.6	18.9	39.6	15.9	38.1	2.5	10	0.599

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.5	76.2	61.5	72.2	59.6	68.9	58.2	63.7	72.5	61.7	69.1	60.0	66.5	5.3	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
60.0	77.3	66.6	58.4	72.9	65.0	56.9	68.9	63.1	29.0	72.8	27.6	69.0	26.4	66.7	74.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.7	11.1	9.1	DB	-1.5	82.8	6.9	4.1	-6.4	85.8	-10.5	88.2	-14.4	90.6	-19.4	93.6	(4)
(5)			WB	-3.8	65.9	6.3	2.0	-8.3	67.3	-12.0	68.5	-15.5	69.6	-20.1	71.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	42.4	27.1	27.9	34.3	40.5	48.7	55.0	59.7	57.2	51.1	42.2	34.3	29.5
	DBStd	13.13	9.52	9.53	5.87	5.35	5.63	4.32	5.02	4.11	4.84	5.96	7.41	9.25
	HDD50	3613	709	619	488	291	94	7	1	2	44	250	472	636
	HDD65	8278	1174	1039	953	736	507	299	179	244	418	707	921	1101
	CDD50	825	0	0	0	5	52	158	301	224	76	9	0	0
	CDD65	16	0	0	0	0	0	1	14	1	0	0	0	0
	CDH74	217	0	0	0	0	3	50	151	13	0	0	0	0
	CDH80	31	0	0	0	0	0	4	27	0	0	0	0	0
Wind	WSAvg	3.2	3.1	3.0	3.4	3.4	4.0	3.7	3.3	2.8	2.7	2.8	2.9	3.2
Precipitation	PrecAvg	61.1	7.2	6.2	5.9	3.5	2.6	3.1	3.3	4.0	5.6	6.2	6.5	7.3
	PrecMax	89.8	15.6	14.1	20.7	6.1	5.2	8.2	5.9	7.4	10.6	15.4	12.2	15.5
	PrecMin	33.1	0.0	0.2	0.6	1.6	0.3	0.7	0.8	0.9	1.2	1.3	0.8	0.6
	PrecStd	12.4	4.3	3.9	4.3	1.3	1.3	1.9	1.2	1.7	2.8	3.5	3.5	4.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.1	48.3	49.3	65.2	73.3	79.8	83.5	75.5	68.4	63.9	54.0	49.5
		MCWB	37.4	39.2	41.7	51.8	57.7	60.7	66.0	60.1	57.0	52.9	45.4	40.9
	2%	DB	40.6	43.6	46.5	60.4	69.5	74.1	78.3	70.8	64.3	56.0	49.4	45.3
		MCWB	34.8	36.9	39.8	48.3	54.8	58.4	63.7	60.6	55.4	48.6	43.4	38.8
	5%	DB	37.9	40.4	43.9	55.2	64.6	69.9	73.6	67.8	61.6	52.9	46.2	41.4
		MCWB	33.9	35.6	37.9	45.2	52.6	56.0	61.9	59.4	54.4	47.8	43.0	37.9
	10%	DB	35.9	37.5	41.2	50.2	60.2	66.1	69.8	65.2	58.7	50.3	42.9	38.7
		MCWB	33.2	35.0	36.4	42.5	50.0	54.5	60.6	58.3	54.0	46.7	40.5	36.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	38.5	41.0	43.8	53.0	59.0	61.9	68.4	64.3	59.6	54.3	47.4	42.2
		MCDB	42.2	46.6	47.7	62.7	70.8	75.0	79.8	71.4	64.0	60.7	50.8	47.0
	2%	WB	36.3	38.2	40.2	48.9	56.1	59.3	64.9	62.0	57.7	51.3	45.3	39.9
		MCDB	38.8	41.8	44.8	58.5	66.1	71.4	74.8	68.2	61.7	55.0	48.1	43.0
	5%	WB	34.9	36.3	38.6	46.1	53.7	57.4	63.1	60.6	55.9	48.9	42.8	38.4
		MCDB	36.7	39.2	42.9	54.0	62.2	66.3	71.7	65.8	59.5	51.6	45.6	41.0
	10%	WB	33.8	35.0	37.2	43.5	51.4	55.9	61.3	59.1	54.4	47.1	40.6	36.8
		MCDB	35.0	37.1	40.4	49.3	59.3	63.1	68.5	64.0	57.7	49.5	43.1	38.9
Mean Daily Temperature Range	5% DB	MDBR	8.9	11.1	13.1	17.0	18.3	19.0	17.5	14.9	13.3	11.5	8.4	8.2
		MCDBR	10.0	11.3	15.5	25.6	28.2	30.1	26.3	20.6	18.2	14.8	13.2	11.4
		MCWBR	6.3	6.6	9.6	14.2	15.0	15.1	13.1	11.5	10.9	9.6	7.7	7.0
	5% WB	MCDBR	8.4	9.7	13.5	22.6	25.0	24.2	24.4	17.5	15.9	12.3	11.5	9.8
MCWBR		5.8	6.1	8.9	13.1	13.9	13.2	13.1	10.6	10.3	8.8	7.6	6.6	
Clear-Sky Solar Irradiance	taub	0.237	0.284	0.319	0.353	0.357	0.356	0.369	0.360	0.332	0.308	0.249	0.207	
	taud	2.065	2.181	2.153	2.137	2.255	2.382	2.382	2.402	2.470	2.393	2.127	1.961	
	Ebn at Noon	150	217	249	263	273	276	269	261	249	211	143	101	
	Edh at Noon	12	22	31	39	38	34	33	30	23	18	11	8	
All-Sky Solar Radiation	RadAvg	65	225	603	1146	1518	1546	1465	1116	691	316	96	42	
	RadStd	15	65	127	194	219	165	158	114	89	47	16	6	

Historical Trends

	DBAvg	Heating		Cooling
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